

Z Gerontol Geriat 2018 · 51:875–881  
<https://doi.org/10.1007/s00391-017-1326-z>  
 Received: 16 October 2016  
 Revised: 17 September 2017  
 Accepted: 21 September 2017  
 Published online: 17 October 2017  
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# Combined treatment with alendronate and *Drynaria* rhizome extracts

## Effect on fracture healing in osteoporotic rats

### Introduction

Osteoporosis is characterized by low bone mass and microarchitectural abnormalities that together increase the fragility fracture risk and will get worse with the gradual aging of the world population [1]. It is expected that 54 million individuals are at risk for low trauma fractures associated with osteopenia or osteoporosis, and the current rates of hip and other fractures in men and women >50 years of age will increase by 50% by 2025 in the USA [2, 3]. Osteoporotic fractures are associated with functional limitations and a decreased quality of life as well as an increase in mortality [4, 5]. According to previous research, osteoporotic bone shows a prolonged and impaired healing process compared with normal bone [6, 7]. The question arises as to whether substances, which are successful in the treatment of osteoporosis, are also efficient in improving fracture healing in osteoporotic bone as well. Recently, there has been heightened interest in using inhibitors of bone resorption for osteoporosis treatment as they are selectively incorporated into osteoclasts and interfere with the resorptive action of osteoclasts [8]. Controlled

trials have shown that recombinant alendronate (ALN) may play a valuable role in the treatment of fractures [9–11]. Due to the effectiveness of oriental medicines on low back pain, the traditional Chinese medicine *Drynaria* rhizome extract (DRE) has been suggested for curing osteoporosis [12]. Previously, it was shown that DRE prevented the development of bone loss induced by ovariectomy in rats, and preserved the fine particle surface of the trabecular bone [13]. In fracture repair, restoration of stability and strength to the injured bone is critical for enabling the affected individual to resume activities of daily living. Consequently, substantial interest exists in accelerating fracture healing and in decreasing the incidence of non-union; however, both antiresorptive and anabolic compounds seem to be oversimplified, and this leads to the plausible hypothesis that combined use of anabolic ALN and antiresorptive DRE may offer significant promise in this respect.

The purpose of this study was to observe the effects of combined treatment with ALN and DRE on fracture healing in osteoporotic rats.

### Material and methods

#### Experimental animals

In this study 70 female Sprague Dawley (SD) rats, aged 5 months with a body weight of 240–260 g, were included. The animals were kept in groups of four per cage with controlled climate conditions (25 °C, 55% humidity, 12 h of light alternating with 12 h of darkness). Free access to tap water and standard laboratory diet containing 1.56% calcium, 0.8% phosphorus, and 800 IU/kg body weight vitamin D were permitted. All the animal experiments were conducted in accordance with international standards on animal welfare as well as being compliant with the Animal Research Committee of the university.

#### Establishment of osteoporotic model

After acclimatization for 1 week, the rats were randomly divided into two groups, including a normal control group ( $N = 5$ ) and an osteoporosis model group ( $N = 65$ ). All rats except those in the normal control group received intragastric administration of retinoic acid (80 mg/kg body weight/day) daily for 2 weeks to induce the osteoporosis model [14]. Af-

Long Chen and Zhou-Shan Tao are co-authors and contributed equally to this work.

ter intragastric administration of retinoic acid, five randomly selected osteoporosis model group rats and the five normal control group ones were euthanized, and the tibial metaphysis was harvested for bone marrow density (BMD) evaluation to confirm the establishment of an osteoporotic animal model.

### Fracture surgery

All rats in the osteoporosis group underwent unilateral transverse osteotomy of the right femur. Briefly, a transverse osteotomy was made at the middle of the right femur, blunt dissections were made along the medial side of the patellar tendon and the patellar bone was dislocated to expose the articular surface of the femoral condyle. A sterilized Kirschner wire (1.0 mm in diameter) was inserted into the femoral medullary canal through the intercondylar notch, and the proximal side of the femur was perforated. The Kirschner wire was bent at the proximal femur to prevent dislocation and cut at the articular surface of the distal femur to allow free joint movement. X-ray analysis (RadSpeed M CH with FPD; Shimadzu, Town, China) was performed to assess the location and quality of fractures immediately after surgery. All the animals received an intramuscular antibiotic and analgesia injection for 3 postoperative days. Unrestricted activity was allowed after the rats recovered from anesthesia.

### Pharmaceutical treatment

After the fracture operation all animals were randomly divided into four groups, and the pharmaceutical interventions commenced from the first postoperative day through the following scheme: vehicle (control), ALN (ALCON, São Paulo, Brazil) 40 mg/kg weekly (group ALN), DRE (Beijing Pharmaceutical, Qihuang, China) 90 mg/kg/day (group DRE), ALN 40 mg/kg weekly plus DRE 90 mg/kg/day (group ALN + DRE). During treatment, the body weight of the animals was measured weekly, and the drug dosage was adjusted accordingly.

### Animal sacrifice and specimen collection

At the end of the observation time (2, 4 or 6 weeks after fracture), animals were euthanized by cardiac puncture under general anesthesia in the early morning. After complete excision of soft tissues, the right femurs were dissected and fixed in 10% formalin solution.

### X-ray analysis

At 2, 4, and 6 weeks post-surgery, conventional anteroposterior and lateral view plain radiographs of the rat femurs were performed using an X-ray imaging system for small animal studies (12 s, 26 kV; Faxitron MX20, Lincolnshire, IL, USA). Bone healing progress and periosteal callus formation were evaluated.

### Histology

At every time point after X-ray analysis, the bones were fixed in 4% paraformaldehyde/phosphate-buffered saline (PBS) overnight at 48 °C and then immediately decalcified with formic acid for 48 h. Decalcified specimens were dehydrated in an ethanol series (80%, 85%, 80%, 90%, 95%, and 100%), embedded in paraffin, and cut into 5-mm thick sections, stained with hematoxylin and eosin (H&E) and qualitatively evaluated by light microscopy (Zeiss Aixoplan with Spot RT digital camera, Zeiss, Jena, Germany). The image was quantitatively analyzed by Image Pro Plus Software (Media Cybernetics, Silver Spring, MD, USA). The total sagittal cross-sectional area (T.Ar; mm<sup>2</sup>), callus area (Cl.Ar; mm<sup>2</sup>), and callus thickness (Cl.Th; µm) were measured at 100× magnification as previously described [15]. These measurements were used to calculate the percentage callus area (%Cl.Ar = Cl.Ar × 100/T.Ar; %).

### Biomechanical testing

The mechanical characteristics of the healing fractures were assessed by a 3-point bending test (Instron 4302, Instron, Norwood, MA, USA) performed at 4 and 6 weeks. The medullary nail was

removed and the bones were preserved at −70° before mechanical testing. Briefly, the femur was placed on two lower support bars 1.8 cm apart in an identical position and the central part of the callus was defined as the testing area. The maximum load and displacement of the maximum load to the breaking point were measured by slowly applying an increased force (loading velocity 2 mm/min, measuring accuracy 0.01 N) to the femoral anterior surface at the fracture site, and the strength of the bones was interpreted as previously described [16].

### Statistical analysis

Data were expressed as mean ± standard deviation (SD). Statistical analyses were performed using the statistics package SPSS 19.0 (SPSS, Chicago, IL, USA). Multiple comparisons between groups were carried out using one-way ANOVA and Tukey's post hoc test. The significance level of 0.05 was applied for all analyses.

## Results

### Animals and confirmation of osteoporosis

In total two rats were excluded from analysis due to an accident in anesthesia, infection, and there were two animals left for evaluation in each group at each observation time. The BMD of the tibial metaphysis was measured in vivo by dual-energy X-ray absorptiometry (Lunar Prodigy Advance, GE Lunar, Madison, WI, USA). The mean BMD of the tibial metaphysis from the normal control and osteoporosis model groups were  $220.47 \pm 30.42$  mg/cm<sup>2</sup> and  $168.21 \pm 26.37$  mg/cm<sup>2</sup>, respectively. In quantitative analysis, the BMD of the tibial metaphysis from control group rats were 31.17% higher than osteoporosis model group animals ( $P < 0.05$ ). These results indicated the successful establishment of osteoporosis in the osteoporosis model group rats.

## Bone X-ray analysis

At 2 weeks after intramedullary nail fixation, anteroposterior radiographs of the fractured femurs showed a transparent zone in the osteotomy gap, which indicated that no ossified calluses formed at that time. Compared to the control group, a wider translucent zone was evident between the broken ends of the right femurs in the DRE, ALN, ALN + DRE groups. At 4 weeks, radiographs revealed external callus production in the ALN + DRE group, and the fracture line had become blurred; however, the osteotomy gap was found in the control group, and the translucent zone was even more obvious. When the fracture line had blurred completely in the control group at 6 weeks post-surgery, the ALN + DRE group showed a continuous reduction in size in radiographs with no evident bridging calluses. In the majority of femurs in the control group, radiolucent zones were still observed between the broken ends of the fractured femurs at 6 weeks. The cases displaying these zones also showed increased radial shortening during fracture healing. In fact, the ALN + DRE group showed proper healing and the bone calluses were undergoing healing and almost achieved bony union; however, a residual gap was still visible in the control group, which brought about bone resorption and even deformity (■ Fig. 1).

## Histological analysis

All groups revealed fibrous callus connection 2 weeks after surgery. The circumference of the fibrous calluses from DRE was larger than that of the control, ALN and ALN + DRE. Besides the abundance of cartilage cells, a small amount of calcification formation was also observed on the control side; however, few cartilage cells were observed along the margin of the fractured femurs from the control and ALN groups (■ Fig. 2) but cartilage tissue and endochondral ossification derived from cartilage cells were pronounced in the fibrous callus zones from DRE (■ Fig. 2). At 4 weeks, numerous mature cartilage cells and calcified tissues were observed in all groups. The majority of fibrous calluses were calci-

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## Combined treatment with alendronate and *Drynaria* rhizome extracts. Effect on fracture healing in osteoporotic rats

### Abstract

The effects of both alendronate (ALN) and *Drynaria* rhizome extracts (DRE) alone could promote bone healing in osteoporotic fractures but there are no reports about the combined use of ALN and DRE for promotion of bone healing of fractures in osteoporotic settings. This study investigated the effects of ALN plus DRE on fractures in osteopenic rats. Osteopenic rats underwent unilateral transverse osteotomy on the femur fixed by a sterilized Kirschner wire 2 weeks after intragastric administration of retinoic acid (80 mg/kg body weight/day). Subsequently, the animals were randomly divided into four groups: control, ALN, DRE and ALN + DRE. All rats from groups ALN, DRE and ALN +

DRE received ALN (40 mg/kg, weekly), DRE (90 mg/kg/day), or both for 2, 4 and 6 weeks. The results of our study indicated that all treatment promoted fracture healing and callus formation compared to controls but ALN + DRE treatment showed significantly stronger effects than ALN or DRE alone in histological, X-ray and biomechanical tests. These results seem to indicate that combined treatment with ALN and DRE has an additive effect on fracture healing and callus formation in osteoporotic rats.

### Keywords

Animals · Plant extracts · Osteoclasts · Traditional Chinese Medicine · X-ray

## Kombinationstherapie mit Alendronat und *Drynaria*-rhizome-Extrakt. Effekt auf die Frakturheilung bei osteoporotischen Ratten

### Zusammenfassung

Die Auswirkungen von sowohl Alendronat (ALN) als auch *Drynaria*-rhizome-Extrakten (DRE) als Einzeltherapie fördern die Knochenheilung von osteoporotischen Frakturen, jedoch existieren keine Berichte über die kombinierte Anwendung von ALN und DRE zur Förderung der Knochenheilung bei Frakturen osteoporotischen Ursprungs. Diese Studie untersuchte die Auswirkungen von ALN plus DRE auf Frakturen bei Ratten mit Osteoporose. Bei osteoporotischen Ratten erfolgte 2 Wochen nach einer intragastralen Gabe von Retinsäure (80 mg/kg Körpergewicht/Tag) eine unilaterale transversale Osteotomie des Femurs, der mit einem Kirschner-Draht fixiert wurde. Anschließend wurden die Tiere randomisiert in 4 Gruppen eingeteilt: Kontrollgruppe, ALN, DRE und ALN + DRE. Alle Ratten der Gruppen ALN, DRE und ALN + DRE erhielten ALN

(40 mg/kg, wöchentlich), DRE (90 mg/kg/Tag) oder beides für 2, 4 und 6 Wochen. Die Ergebnisse unserer Studie deuteten darauf hin, dass alle Behandlungen im Vergleich zur Kontrollgruppe die Frakturheilung und Kallusbildung förderten, wobei die Therapie mit ALN + DRE in histologischen, radiologischen und biomechanischen Untersuchungen signifikant stärkere Auswirkungen zeigte als ALN oder DRE alleine. Diese Ergebnisse scheinen darauf hinzuweisen, dass eine Kombinationstherapie mit ALN und DRE einen zusätzlichen Effekt auf die Frakturheilung und Kallusbildung bei osteoporotischen Ratten hat.

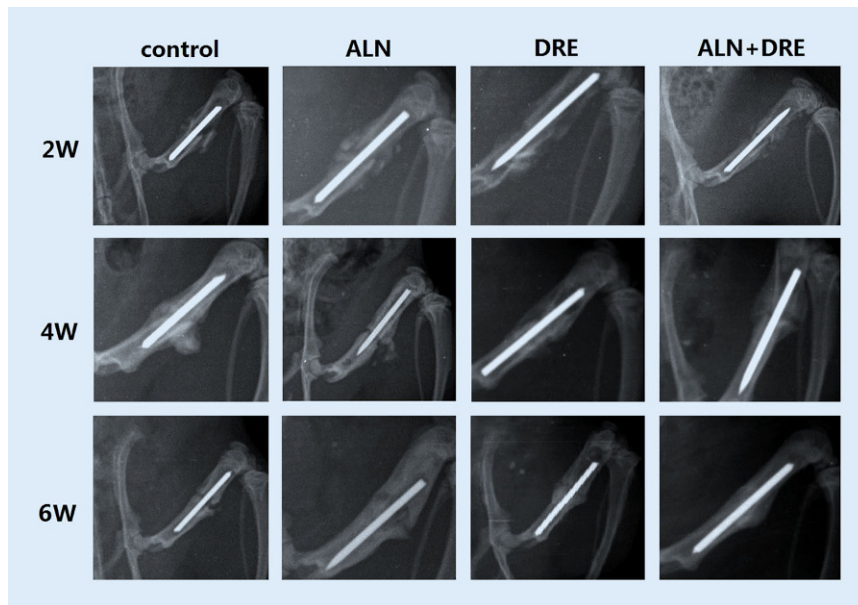
### Schlüsselwörter

Tiere · Pflanzenextrakte · Osteoklasten · Traditionelle Chinesische Medizin · Röntgen

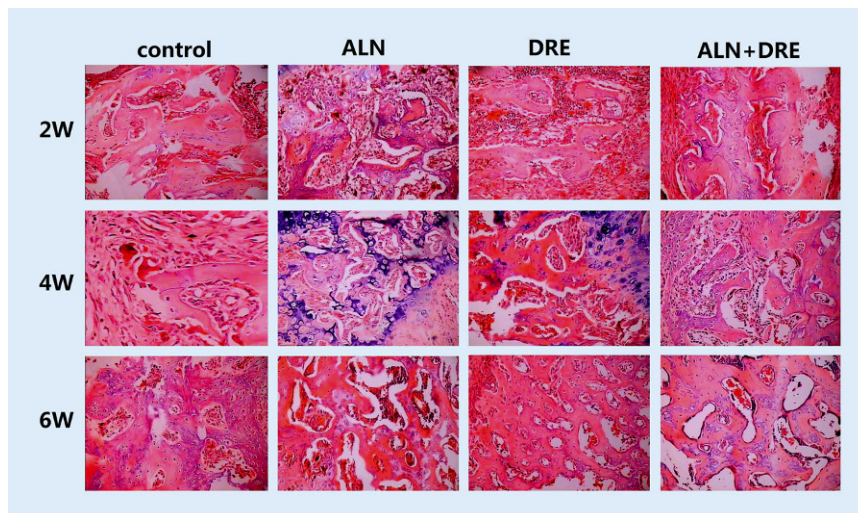
fied into osseous calluses, which grew within the medullary cavity of femurs as internal callus unions; however, neither osseous calluses nor internal calluses were detected from controls (■ Fig. 2). At 6 weeks, there appeared to be even more osseous calluses than observed at 4 weeks from all groups. A large number of newly formed blood vessels along the medullar canals were observed. In addition, the

femur osteotomy site was surrounded by newly formed bone that consisted of woven bone and primary bone union; however, osseous calluses were rare, and the specimens showed a more advanced stage of bone resorption from control (■ Fig. 2). Fibrous calluses surrounding the ends of fractured femurs were completely calcified into woven bone from ALN, DRE and ALN + DRE. The quantity of new





**Fig. 1** ▲ Radiographs of the healing fractures of all groups at 2, 4 and 6 weeks post-surgery. W weeks, ALN alendronate, DRE *Drynaria* rhizome extracts



**Fig. 2** ▲ Decalcified bone histology at 2, 4 and 6 weeks post-surgery (H&E 100 × magnification). W weeks, ALN alendronate, DRE *Drynaria* rhizome extracts

bone was greater than that observed at 4 weeks and the calcified bone calluses had undergone remodelling and consolidation.

### Quantitative analysis of the fracture calluses

Quantitative analysis showed that at 2, 4 and 6 weeks post-fracture, the T.Ar of the femur was larger in the DRE and the ALN + DRE groups compared with the ALN and control groups ( $P < 0.01$ ) (■ Fig. 3a). Furthermore, at 4 and 6 weeks

post-fracture, the T.Ar of the femur in the ALN, DRE and ALN + DRE groups was higher than that in control ( $P < 0.001$ ) (■ Fig. 3a). The Cl.Th and percentage callus area were also larger in the ALN groups compared with the control group at both end points ( $P < 0.01$ ) (■ Fig. 3b, c). In addition, T.Ar, Cl.Th and percentage callus area in the ALN + DRE group were the highest among the four groups.

### Biomechanical test

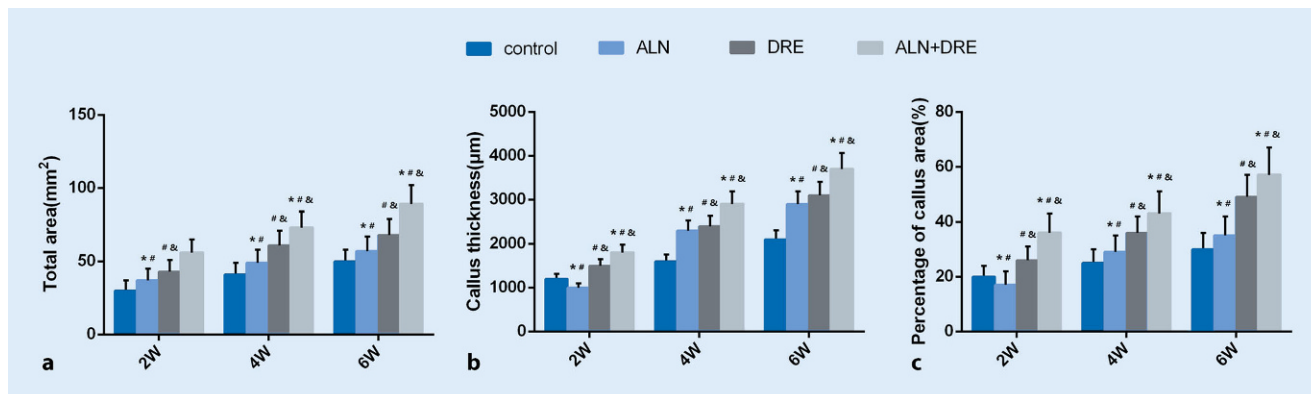
Results of biomechanical testing of fractured femurs were expressed as ultimate load and stiffness (■ Fig. 4b). From 4–8 weeks post-fracture, the biomechanical strength increased with time in all groups. At each observation time, the strongest effects on strength of fractured femurs were observed in the ALN + DRE group ( $P < 0.05$ ). Compared to the control, ALN, DRE and ALN + DRE increased the ultimate load by 2.8, 3.0 and 3.8-fold at 4 weeks and 2.17, 2.44 and 3.66-fold at 8 weeks and the stiffness by 1.7, 1.8 and 2.1-fold at 4 weeks and 1.36, 1.43 and 1.79-fold at 8 weeks post-fracture, respectively ( $P < 0.05$ ).

### Discussion

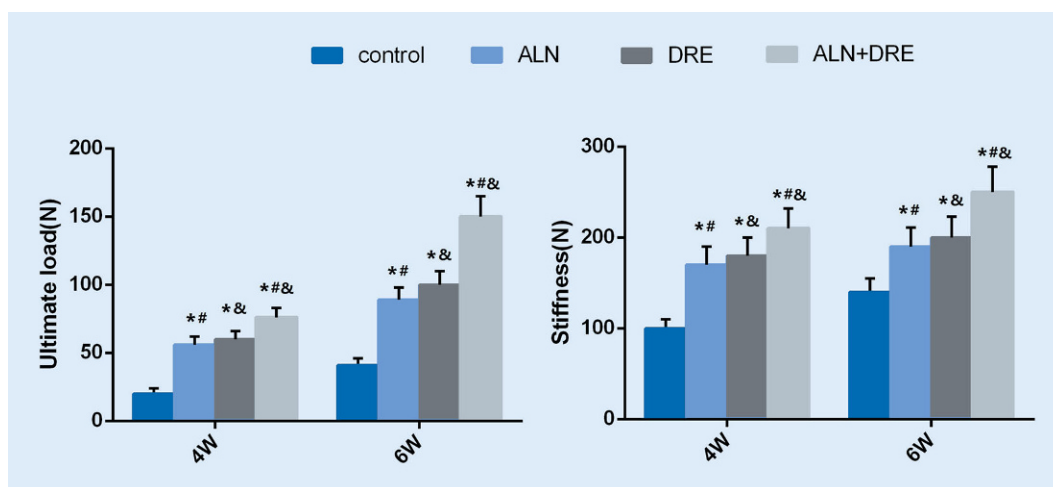
In this study, ALN + DRE showed stronger effects on fracture healing than either monotherapy in osteoporotic rats, with the highest values of callus area, callus thickness and the percentage callus area (2, 4 and 6 weeks) in histological evaluation, the enhanced fracture healing in bone X-ray analysis, and the strongest effects on callus strength in biomechanical testing. Moreover, ALN showed no positive effects on fracture healing than control groups in assessments of the fractured site by X-ray, histology, and biomechanical testing at early stages (2 weeks). These results seemed to indicate that combined treatment with ALN and DRE had an additive effect on fracture healing and callus formation in osteoporotic rats.

In the experiments conducted, SD rats were used as an animal model by intragastric administration of retinoic acid. Osteoporosis in the animal model was confirmed by dual-energy X-ray absorptiometry. The proximal femur was chosen as the fracture site because it is a clinically relevant region representing a frequently fractured area in osteoporotic patients and after the model is successfully developed parameters of all bones change and the change of femur is most obvious [17]. Because of this, the femur was selected as the experimental object.

The bone is under constant remodelling, with repeated cycles of bone



**Fig. 3** ▲ Quantitative analysis of the fracture calluses at 2 weeks (2W), 4 weeks (4W) and 6 weeks (6W) using H&E staining (a, b, and c)  $N = 4$  specimens/group. Data are expressed as mean  $\pm$  SD; error bars in the figure indicate SD. \* $P < 0.05$  vs. ALN, # $P < 0.05$  vs. DRE, & $P < 0.05$  vs. 10% ALN + DRE (by one-way ANOVA and Tukey's post hoc test). W weeks, ALN alendronate, DRE *Drynaria* rhizome extracts



**Fig. 4** ◀ Biomechanical results at 4 and 6 weeks expressed as ultimate load force and stiffness ( $N = 4$  specimens/group). Data are expressed as mean  $\pm$  SD, error bars in the figure indicate SD. \* $P < 0.05$  vs. ALN, # $P < 0.05$  vs. DRE, & $P < 0.05$  vs. 10% ALN + DRE (by one-way ANOVA and Tukey's post hoc test). W weeks, ALN alendronate, DRE *Drynaria* rhizome extracts

resorption by osteoclasts followed by deposition of new bone by osteoblasts. This process ensures the repairing of microfractures as well as remodeling the bony architecture in response to stress [18]. Previous studies indicated that osteoporosis occurs as a result of a persistent decrease in bone formation and a continued increase of bone resorption [19]. To date, clinical and experimental studies have consistently demonstrated that bone healing in postmenopausal osteoporotic women and estrogen depletion-induced osteoporotic animals is substantially delayed or impaired [7, 20, 21]. Alendronate, which is commonly used in the treatment of postmenopausal osteoporosis, is a potent inhibitor of osteoclast-induced bone resorption. The reduction in the risk of fracture during antiresorptive treatment has been

related to the magnitude of changes in bone mineral density and remodelling activity [22, 23]. The histological, radiology and biomechanics findings of the current study showed the fracture healing in the bisphosphonate group to be significantly advanced compared to the control group at 4 and 6 weeks.

Screening traditional Chinese medical herbs for effective drugs that can suppress bone resorption and promote bone regeneration is one of the ways to discover new treatment for osteoporotic fractures. The DRE is a traditional Chinese herb well known for its effects of promoting patient latent self-repair mechanisms and bone healing [24, 25]. The in vitro study demonstrated that DRE could enhance the proliferation and differentiation of MC3T3-E1 cells [26]. Research on DRE extracts showed DRE-stimulated prolifer-

ation and differentiation of some osteoblastic cell lines [27]. Furthermore, it was observed to inhibit osteoclast formation [27]. In our study the results indicated that it could increase the effect on fracture healing and callus formation in osteoporotic rats.

In our study, combined therapy with ALN and DRE induced an additive effect on fracture healing in osteoporotic rats. Previous studies have shown that DRE could inhibit osteoclast formation [27] and enhance the proliferation and differentiation of MC3T3-E1 cells [26], which can suppress bone resorption and promote bone regeneration. Fracture healing progresses in four stages: inflammation, soft callus formation, hard callus formation and remodelling, with some degree of overlap of the stages [28]. Bone remodelling occurs in conjunction with

cartilage mineralization, vascular invasion and formation of woven bone. Additionally, osteoclasts are active from the very early phases of fracture healing removing necrotic bone from the fracture site [29]. So Alendronate sodium showed no more positive effects on fracture healing than control groups in assessment of the fractured site at 2 weeks for inhibiting osteoclast function. When at 4 and 6 weeks, ALN could suppress bone resorption by osteoclasts, formation of osteoclasts or interaction between osteoclasts and osteoblasts is not prevented. At the same time, therapy with DRE could enhance the function of ALN for inhibiting osteoclasts at 2 weeks.

Our study had several limitations. Firstly, the number of animals is limited due to the preclinical pilot study design. In addition, there was no group after 6 weeks; however, the primary aim of this pilot study was to investigate the potential of early new bone formation applied to fractures. Secondly, the doses of ALN and DRE were determined according to experiments on osteoporotic fractures, which can affect the results of our study.

## Conclusion

In summary, our study suggests that combined treatment with ALN and DRE has an additive effect on fracture healing and callus formation in osteoporotic rats. More research will be necessary to prove the advantage of administration of ALN in combination with DRE on fracture healing.

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**Funding.** This work was funded by a research grant to Zhejiang Province Medicine and Health Research Project (No. 2016KYB196), Zhejiang Chinese Medicine Bureau Research Project (No. 2016ZA140), Science Technology Department of Zhejiang Province Research Project (No. 2017C37125) and Key Project of Natural Science Research for

Colleges and universities of Anhui Province Project (No.KJ2017A266).

## Compliance with ethical guidelines

**Conflict of interest.** L. Chen, Z.-S. Tao, H. Chen, K. Zhou and D.-S. Zhou declare that they have no competing interests.

All animal experiments were conducted in accordance with international standards on animal welfare as well as being compliant with the Animal Research Committee of the university.

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## Gesetzlich nicht normierte ärztliche Auskunfts- und Offenbarungspflichten

MedR Schriftenreihe Medizinrecht

**Berlin: Springer-Verlag GmbH Germany 2017, 1. Aufl., 308 S., (ISBN: 978-3662537985), Taschenbuch 74,99 EUR**

Mit dieser juristischen Promotionsarbeit aus dem Jahr 2013 hat sich der Autor auf 195 Seiten intensiv mit Fragen zur umstrittenen Rechtsfigur der medizinischen Grundaufklärung befasst. Zutreffend wird bereits zu Beginn (S. 3) bemerkt, dass der Gesetzgeber die Grundaufklärung im Zuge der Gesetzesreform durch das Patientenrechtegesetz mit keinem Wort erwähnt hat (siehe auch BR-Drs. 312/12 sowie BT-Drs. 17/11710 und 17/10488). Im weiteren Verlauf des Buches (z. B. Aufklärung über Behandlungsalternativen (S. 109 ff.) oder zur Sicherungsaufklärung (S. 175 ff.)) wird aber auf die neuen Vorgaben im BGB, z. B. Aufklärungspflichten über Behandlungsalternativen nach § 630e Abs. 1 S. 3 BGB oder die Informationspflichten nach § 630c Abs. 2 S. 1 BGB, nicht mehr eingegangen.

Dezidiert führt der Autor die vielfältigen Begriffe im Kontext der Aufklärung an und grenzt die Grundaufklärung zur Aufklärung im Großen und Ganzen ab. Zudem un-

ternimmt er einen Definitionsversuch zur Grundaufklärung, bei der er weder die Dringlichkeit noch Behandlungsalternativen berücksichtigen möchte und zudem auf eine „allgemein verständliche Wissensvermittlung im Großen und Ganzen“ (S. 161) abstellt. Aufgrund der Vorgaben im BGB und den Ausführungen in den Gesetzesmaterialien zum Patientenrechtegesetz dürfte sich der Definitionsversuch von Schenk zur Rechtsfigur der medizinischen Grundaufklärung in der Praxis kaum durchsetzen können. Hier ist auf die nicht enumerativen Angaben zu den Aufklärungs- und Informationspflichten im Gesetzeswortlaut zu verweisen (z. B. §§ 630c Abs. 2 S. 1, 630e Abs. 1 S. 1, 2 BGB). Für die Aktualität des Buches (Erscheinungsjahr 2015) wäre es sicherlich hilfreich gewesen, die Änderungen durch das Patientenrechtegesetz aus dem Jahr 2013 stärker einzubeziehen. Hierbei ist wohl zu berück-

sichtigen, dass es zur sicherlich unglücklichen Überschneidung mit dem Abschluss des Promotionsvorhabens und der Gesetzesreform gekommen ist. Das Buch von Schenk eignet sich für akademische teilweise historische Diskussionen über unterschiedliche Aspekte der (Grund-) Aufklärung, empfiehlt sich gerade wegen der fehlenden Aktualität im Hinblick auf die Änderungen durch das Patientenrechtegesetz für den praktisch tätigen Rechtsmediziner aber nicht.

**Markus Parzeller, Frankfurt**

